BUSINESS REQUIREMENTS DOCUMENT

PROJECTDETAILS

PROJECT NAME	DOCUMENT #
Telecom Customer Churn Analysis	BRD-TELCO-2025-001
CREATOR	**DATE**
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VERSION	
1	

1. EXECUTIVE SUMMARY

This BusinessRequirements Document (BRD) provides a clear and simple outline of the business needs for a churn prediction system in the JB Link telecom company. The main goal is to identify which customers are likely to leave and take timely action to retain them. The project will involve analyzing data related to customer behavior, usage, and satisfaction to predict churn, understand key issues, and improve customer retention.

We aim to build a set of dashboards that will help monitor churn trends, understand why customers leave, and identify the segments that need special attention. The BRD includes the current state of churn analysis, the proposed improvements, business drivers, key deliverables, and the specific functional requirements.

This document is intended for stakeholders such as business analysts, data analysts, BI developers, product managers, and decision-makers who are involved in planning, implementing, or using churn-related insights for strategy and operations.

2. PROJECT DESCRIPTION

Thepurposeof this projectis to developachurn analysis and prediction solution for the telecom company. Currently, customer-related data (usage, satisfaction, complaints) is scattered across systems, making it difficult to identify at-risk customers early. The existing process is manual, slow, and lacks consistency, resulting in missed opportunities to retain valuable customers. This project will address these issues by creating a unified churn model and insightful dashboards to help the company take proactive, data-driven retention actions. By integrating and analyzing data from usage, satisfaction scores, demographics, and complaints, the company can proactively retain at-risk customers.

3. PROJECT SCOPE

Thisprojectwill focusonbuilding a churn analysis solution using data from existing internal sources and presenting insights through Power BI dashboards. The scope includes the development of models, dashboards, and an alerting system but excludes real-time integration or external data enhancements

IN-SCOPE ITEMS	OUT-OF-SCOPE ITEMS
Item 1:Consolidation of customer data from multiple departments (usage, satisfaction, demographic, complaints)	Item 1:Real-time API integration
Item 2: Building Excel-based churn prediction model using scoring and lookup formulas	Item 2:External data enrichment (third-party sources)
Item 3: Creating Power BI dashboards for churn insights, driver analysis, and customer profiling	Item 3:Full machine learning automation
Item 4: Implementing alert system based on churn score and risk levels	Item 4:Mobile app/dashboard version

4. BUSINESS OBJECTIVES

1:	Predictchurnriskaccurately using structured scoring models

2:	Build interactive dashboards to monitor trends, segment risks, and track churn over time
3:	Segment customers to enable targeted retention strategies
4:	Identify customers at high risk of churn based on behavioral, demographic, and usage data

5. PRESENT PROCESS

Requirements Gathering State

Currently, customer complaints, usage history, and churn feedback are stored in system. Churn is analyzed manually at the end of a quarter using powerbi. Customer interviews are not documented in any analytical format. There is no scoring or alert system for early intervention.

6. PROPOSED PROCESS

-**Automated Churn Prediction:** Use scoring rules (e.g., low satisfaction + high charges) to flag at-risk customers.

-**Early Alert System:** Highlight high-risk customers before churn happens to enable proactive retention.

-Structured Feedback Integration: Capture churn reasons from interviews and complaints in a searchable format to guide strategy.

-**Customer Segmentation:** Categorize customers (e.g., High CLTV + High Risk) for targeted campaigns.

-**Visual Dashboards**: Create Power BI dashboards:
Dashboard 1: Customer Demographics
Dashboard 2: Usage Trend Analysis
Dashboard 3: Churn Analysis
Dashboard 4: Retension Analysis

7. FUNCTIONAL REQUIREMENTS

ID	PRIORITY	DESCRIPTION
FR1	High	Consolidate customer data from multiple sources into a unified churn model worksheet
FR2	High	Create Power BI dashboards with KPIs (Total Customers, Churn Rate, CLTV, etc.) Enable
FR3	Medium	drill-down and filter features in dashboards (e.g., by age group, contract type) Create
FR4	Medium	alerts/flags for customers at high risk of churn based on score threshold Include
FR5	Medium	comparison visuals (churned vs. retained) by payment method or city Add ability to
FR6	Low	export dashboard visuals to PDF/Excel for reporting purposes

8. NON-FUNCTIONAL REQUIREMENTS

ID	REQUIREMENT
NFR1	Dashboards must refresh with new data monthly
NFR2	The churn model worksheet must be scalable to 10,000+ customers
NFR3	All visuals must load within 5 seconds in Power BI
NFR4	User access should be role-based (Viewer/Editor)

9. PROJECT CONSTRAINTS

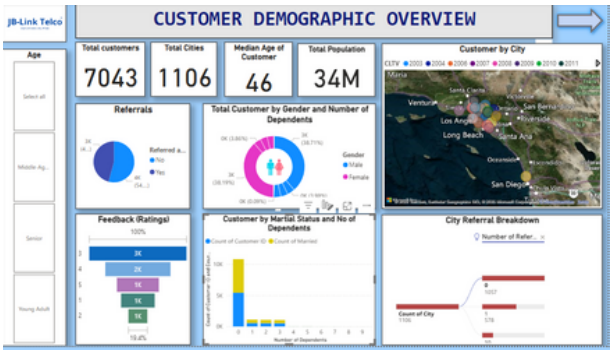
NAME	LOCATION
Data Availability	The project relies on internal data sources; missing or incomplete data may limit insights.
Tool Limitations	Excel and Power BI are the primary tools; no advanced ML models or external APIs are used.
Time Constraint	The project must be completed within a week timeframe using internal resources.

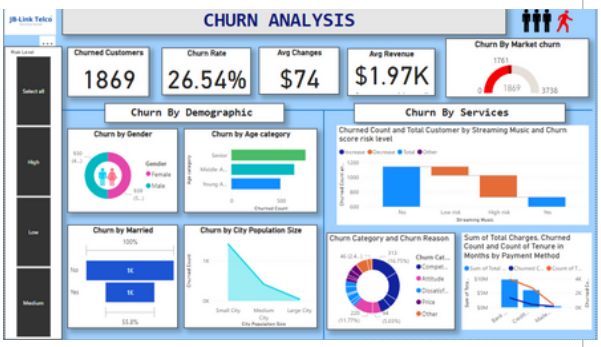
10. GLOSSARY

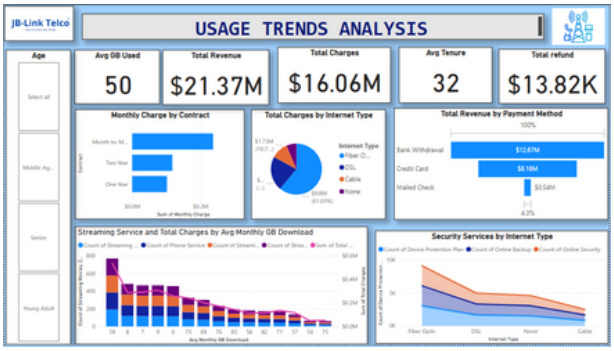
TERM/ABBR	EXPLANATION
CLTV	Customer Lifetime Value
KPI	Key Performance Indicator
BA	Business Analyst
BI	Business Intelligence
Churn	Customer cancelling or discontinuing service

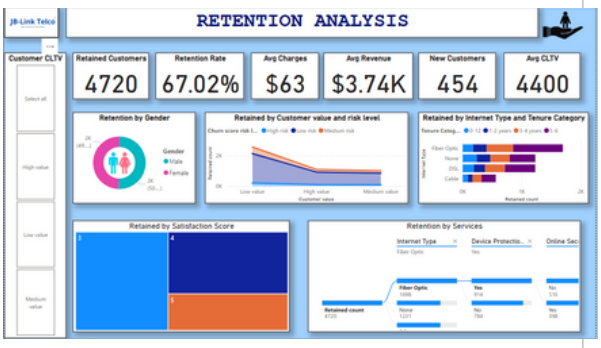
11. APPENDIX

Dashboard









UML Diagrams

ClassDiagram:Shows entities like Customer, Service, Status, and their attributes (e.g., Age, Tenure, ChurnFlag)

Use Case Diagram:Actors: Business Analyst, BI Developer, Executive
Use Cases: View churn dashboard, Analyze churn drivers, Generate retention report

Sequence Diagram:Scenario: Service Call → Complaint Raised → Update Status → Impact Churn Score

Activity Diagram:Steps: Data Collection → Scoring → Dashboard Update → Churn Alert Trigger

Profile Diagram:Segments like "Young Low Tenure High Risk" or "Senior High CLTV Low Risk" with associated behavioral traits